

Practice Test 1 · Answer Key

Physics Midterm · 80 points

Part A · Multiple Choice (2 points each, 34 total)

Quick key: 1 C 2 A 3 D 4 B 5 C 6 B 7 D 8 B 9 C 10 D 11 C 12 D 13 A 14 C 15 B 16 D 17 A

#	Ans	Why
1	C	displacement. Displacement has a size and a direction. The other three have size only.
2	A	distance, speed, mass. Velocity, force and acceleration are vectors, so B, C and D each contain a vector.
3	D	10 N. Perpendicular vectors: $R = \sqrt{(6.0^2 + 8.0^2)} = 10 \text{ N}$. Adding to get 14 N only works for vectors in the same direction.
4	B	17.3 m/s. $v_x = 20 \cos 30^\circ = 17.3 \text{ m/s}$. 10 m/s is the vertical component ($20 \sin 30^\circ$).
5	C	400 m and 0 m. Distance is the path length (400 m). Displacement is start to finish, which is zero.
6	B	4.0 m/s². $a = (v - u)/t = (30 - 10)/5.0 = 4.0 \text{ m/s}^2$.
7	D	acceleration. Slope = $\Delta v/\Delta t$ = acceleration. The area under the line gives the displacement.
8	B	zero and 9.81 m/s² downward. The velocity is momentarily zero, but gravity still acts, so the acceleration is still 9.81 m/s ² downward.
9	C	44.1 m. $s = \frac{1}{2}gt^2 = \frac{1}{2}(9.81)(3.0)^2 = 44.1 \text{ m}$. 29.4 is its speed in m/s (gt), not the distance.
10	D	12 m/s. $v = u + at = 0 + (2.0)(6.0) = 12 \text{ m/s}$. 36 is the distance in meters ($\frac{1}{2}at^2$).
11	C	both land at the same time. Both start with zero vertical velocity from the same height with the same g. Horizontal motion does not change the fall time.
12	D	stays constant. No horizontal force acts, so $a_x = 0$ and v_x does not change.
13	A	zero. Newton's first law: constant velocity means zero net force (balanced forces).
14	C	3.0 m/s². $a = F_{\text{net}}/m = 12/4.0 = 3.0 \text{ m/s}^2$.
15	B	mass stays the same and weight decreases. Mass is the amount of matter and does not change. Weight = mg depends on g.
16	D	equal in size. Newton's third law: the two forces are equal in size and opposite in direction. The car has the larger acceleration because its mass is smaller.
17	A	the book pulling up on the Earth. A third-law pair is the same type of force acting on the two interacting objects: Earth on book and book on Earth. B also acts on the book, so it cannot be the partner.

Part B · Free Response (46 points)

18. Vectors and scalars (10 points)

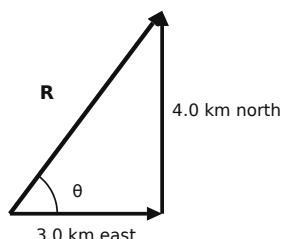
(a) distance **scalar** · displacement **vector** · speed **scalar** · velocity **vector** · mass **scalar** · weight **vector**

½ point each. Weight is a force, so it is a vector (it points down).

(b) $F_x = 50 \cos 37^\circ = \mathbf{39.9 \text{ N}}$ · $F_y = 50 \sin 37^\circ = \mathbf{30.1 \text{ N}}$

1 point each. No credit if sin and cos are swapped or the unit is missing.

(c)



$R = \sqrt{(3.0^2 + 4.0^2)} = \mathbf{5.0 \text{ km}}$. $\theta = \tan^{-1}(4.0/3.0) = \mathbf{53.1^\circ \text{ north of east}}$.

Diagram: 1 point for the second arrow starting at the tip of the first, 1 point for the resultant drawn from the start to the finish with an arrowhead. Calculation: 1 point Pythagoras, 1 point 5.0 km, 1 point for the angle with its reference direction ("north of east").

19. Motion in a straight line (10 points)

(a) $a = \text{slope} = (12 - 0)/(4 - 0) = \mathbf{3.0 \text{ m/s}^2}$

(b) The cyclist moves at a **constant velocity** of 12 m/s (acceleration zero). The **net force is zero**.

1 point each. "Not moving" earns no credit: the line is flat at 12 m/s, not at zero.

(c) $a = (0 - 12)/(14 - 10) = \mathbf{-3.0 \text{ m/s}^2}$ (slowing down)

1 point for the slope calculation, 1 point for the negative sign.

(d) Displacement = area under the graph = $\frac{1}{2}(4)(12) + (6)(12) + \frac{1}{2}(4)(12) = 24 + 72 + 24 = \mathbf{120 \text{ m}}$

1 point for using the area, 1 point for 120 m.

(e) At the top $v = 0$: $t = u/g = 19.6/9.81 = \mathbf{2.00 \text{ s}}$. Height: $H = u^2/(2g) = 19.6^2/(2 \times 9.81) = \mathbf{19.6 \text{ m}}$

1 point each.

20. Projectile motion (10 points)

(a) Vertical motion, $u_y = 0$: $h = \frac{1}{2}gt^2$, so $t = \sqrt{2h/g} = \sqrt{2 \times 30/9.81} = \mathbf{2.47 \text{ s}}$

1 point: $u_y = 0$. 1 point: correct equation. 1 point: answer with unit.

(b) $x = u_x t = 12 \times 2.473 = \mathbf{29.7 \text{ m}}$

(c) $v_y = gt = 9.81 \times 2.473 = \mathbf{24.3 \text{ m/s downward}}$ (or $v_y = \sqrt{2gh} = \sqrt{2 \times 9.81 \times 30}$)

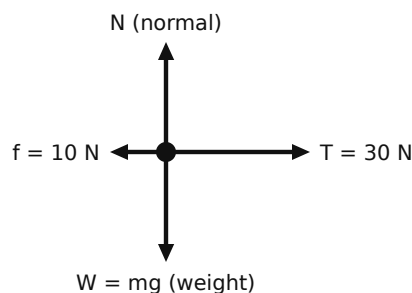
1 point for the value, 1 point for the direction (or a negative sign). If the rounded time 2.47 s was used, accept 29.6 m in (b) and 24.2 m/s in (c).

(d) $v = \sqrt{(12^2 + 24.3^2)} = \mathbf{27.1 \text{ m/s}}$. Angle = $\tan^{-1}(24.3/12) = \mathbf{63.7^\circ}$ below the horizontal

1 point: combining the two components with Pythagoras. 1 point: speed. 1 point: angle.

21. Newton's second law (10 points)

(a)



1 point: weight down and normal up, equal lengths. 1 point: tension to the right. 1 point: friction to the left and shorter than the tension. Subtract 1 point for any extra force (for example "ma" or "force of motion").

(b) $W = mg = 5.0 \times 9.81 = \mathbf{49.1 \text{ N}}$. There is no vertical acceleration, so $N = W = \mathbf{49.1 \text{ N}}$.

(c) $F_{\text{net}} = 30 - 10 = \mathbf{20 \text{ N to the right}}$. $a = F_{\text{net}}/m = 20/5.0 = \mathbf{4.0 \text{ m/s}^2}$ to the right.

1 point: subtracting friction. 1 point: 20 N. 1 point: 4.0 m/s^2 . No credit for $a = 30/5.0 = 6.0 \text{ m/s}^2$.

(d) $v = u + at = 0 + (4.0)(3.0) = \mathbf{12 \text{ m/s}}$. $s = \frac{1}{2}at^2 = \frac{1}{2}(4.0)(3.0)^2 = \mathbf{18 \text{ m}}$.

22. Newton's first and third laws (6 points)

(a) The passenger has **inertia**: a body **at rest stays at rest** unless a net force acts on it. The floor pulls the feet forward with the bus, but the upper body stays where it was, so the passenger falls backward relative to the bus.

1 point: inertia / first law stated. 1 point: no force acts on the upper body, so it stays at rest. No credit for "a force pushes the passenger backward".

(b) Her hands push **backward on the water**; the water pushes **forward on her hands** with an equal force. That forward force acts **on the swimmer**, so it accelerates her forward.

1 point: partner force named (water on swimmer, forward, equal size). 1 point: this force acts on her, so she accelerates.

(c) Both forces act on the **same object** (the book). A third-law pair always acts on **two different objects** and is the same type of force. The partner of the weight is the book pulling up on the Earth; the partner of the normal force is the book pushing down on the table.

1 point: same object, so not a pair. 1 point: one correct partner named. The two forces are equal here only because the book is in equilibrium (first law).